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Contextual Fibration Attention

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Contextual Fibration Attention

Gary Nan Tie, Aug 29, 2025

Abstract In a query-key-value attention model, the key-value pairing conveys language semantics. Query context also carries meaning. We introduce an abductive fibration attention model that holistically captures both. A fibration lifting, jointly learns from query context and language semantics (that abductively finds a key to best explain the returned value).

We adopt the notation from [3]:

key-value pairs (k_j, v_j) , $j \in [m]$

queries $q_1, \dots, q_{i-h}, \dots, q_i, \dots, q_{i+h}, \dots, q_n$

$w = 2h+1$ window, w small, say ≤ 5

Let $g_1, \dots, g_w$ be from a Beta distribution

centered at i , $\sum_{l=1}^w g_l = 1$.

Learn σ_i jointly from the context window via:

$$\begin{array}{ccc} Q^w \otimes K & \xrightarrow{m^w \otimes 1} & (Q')^w \otimes K \\ \downarrow 1^w \otimes p & \sigma_i \dashrightarrow & \downarrow 1^w \otimes p \\ Q^w \otimes V & \xrightarrow{m^w \otimes 1} & (Q')^w \otimes V \end{array}$$

so that the diagram commutes up to $=_\varepsilon$,

where $p: K \rightarrow V$ with $p(k_j) =_\varepsilon v_j$, $j \in [m]$

and for $l \in [w]$, $m^l(q_l) = q_l^0 \triangleq \sum_{j=1}^m \alpha_{l,j}^0 v_j$,

$\alpha_{l,j}^0 = \text{softmax}_j \text{score}(q_l, k_j)$, $j_l^0 \triangleq \arg \max_j \alpha_{l,j}^0$,

score a similarity function, like a reproducing kernel.

$$\text{Let } q_l^1 \triangleq \pi_l \sigma_i \begin{bmatrix} q_i^0 \\ \vdots \\ q_w^0 \\ v_i^0 \end{bmatrix}, l \in [w], \sigma_i: Q^w \otimes V \rightarrow (Q')^w \otimes K$$

Let $\alpha_{l,j}^1 = \text{softmax score}(q_l^1, k_j)$ and $j_l^1 = \arg \max_j \alpha_{l,j}^1$.

There is an l that maximizes $\{q_l \alpha_{l,j_l^1}^1\}_{l=1}^w$, denote this by l^* .

Each query update q_l^1 in a context window centered at i

has a probability distribution $\alpha_{l,j}^1$, $j \in [m]$, over the keys

$k_1, \dots, k_m$. From the product distribution $\{q_l \alpha_{l,j}^1\}_{l=1}^w \prod_{j=1}^m$

we are choosing the most likely l via $q_i^1 \triangleq q_{l^*}^1$

Schematically:

$$\text{input } q_i \mapsto \left. \begin{array}{ccc} q_i^0 & \mapsto & q_i^1 \\ \vdots & & \vdots \\ q_w^0 & \mapsto & q_w^1 \end{array} \right\} q_{l^*}^1 = q_i^1 \text{ output}$$

q_l^0 is grounded by similarity,

q_l^1 is learned from σ_i that takes into account

semantics $p: K \rightarrow V$.

Query refinement process:

- Input query q_i
- Update q_i to q_i^1
- Replace q_i with q_i^1
- Repeat multiple times
- Return refined query $q_{L^*}^1$ and value $v_{j^*}^1 = p(k_{j^*}^1)$

Summary: In a context window around query q_i ,

each shoulder query is initially updated (the endpoint cases miss a shoulder) using a query-key similarity

function, like a reproducing kernel, to create I/O

$(q_L, p(k_j)) / (q_L^0, k_j)$ to learn lifting σ_i .

σ_i is then used to update q_L^0 to q_L^1

This iteration yields q_i^1 from a Beta prior and the process repeated.

Note that σ_i updates both contextually and semantically!

References

[1] 'Multi-head abductive attention'

Gary Nan Tie, Aug 25, 2025

DOI: 10.13140/RG.2.2.20597.23521

Transformer multi-head attention can focus on different parts of an input query sequence, however without regard to potentially different key-value semantics. This new primitive for formal query updating, constructs updates to a query in parallel, by learning multi-head abductive fibration liftings, to refine similarity based attention, utilizing the semantics of multiple key-value pairings.

[2] 'Formal query updating to generate problem specific fibration abduction attention algorithms'

Gary Nan Tie, Aug 23, 2025

DOI: 10.13140/RG.2.2.33029.41446

A formalism to construct problem specific query update algorithms founded on standard attention and fibration attention primitives.

[3] 'FibChat - It makes stuff up, but with structure.'

Gary Nan Tie · July 2025

DOI: 10.13140/RG.2.2.10549.59364/2

FibChat is a LLM based on fibration attention, that enables coherent structured mitigation of hallucinations.

[4] 'Fibrations explain all you need!

- Fibrations, Abduction, Attention

Gary Nan Tie, Mar 4, 2025

DOI: 10.13140/RG.2.2.35984.52488

To conceptualize and perform abductive machine learning, categorical fibrations are proposed, as they coherently give both context and structure for abductive reasoning via an attention mechanism. Moreover, 2-categorical lifting compatibility conditions ensure consistent hierarchical and parallel explanations.